

Analysis First-Year Exam, August 2022, Part 2

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let $f: [0, 1] \rightarrow \mathbb{R}$ satisfy the following rules: if t is irrational then $f(t) = 0$; if $t = m/n$ is a rational in lowest terms then $f(t) = 1/n$. Prove that f is Riemann integrable and calculate its integral.
2. Let $\mathcal{F} \subseteq C(X)$ be equicontinuous and let A be the set comprising all points of X at which $\mathcal{F}$ is bounded; that is, $a \in A$ exactly when there exists K such that $|f(a)| < K$ whenever $f \in \mathcal{F}$. Prove that $A \subseteq X$ is both closed and open.
3. Let the function $f: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable to all orders and denote its n th derivative by f_n . Assume that the sequence $(f_n)_{n=0}^\infty$ converges uniformly to the function g on $\mathbb{R}$ and deduce as much as possible about g .
4. Prove that if the real-valued functions f and g on the same space are measurable, then so are their pointwise sum $f + g$ and pointwise product fg .
5. For each $n \in \mathbb{N}$ let $f_n: [0, 1] \rightarrow [0, 1]$ be continuous and assume that $f_n \rightarrow 0$ pointwise as $n \rightarrow \infty$. Does it follow that

$$\int_0^1 f_n(t) dt \rightarrow 0 \quad \text{as } n \rightarrow \infty?$$

Does your answer change if continuous is replaced by Riemann integrable?